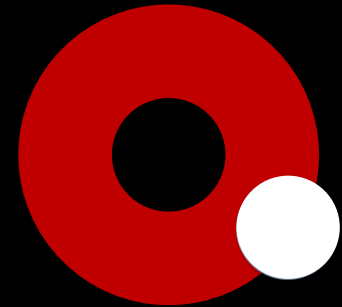




Quantum | nll

分子 | 尼奧

*Foundry of Hybrid Nanoimprint
Lithography Solution for Material
Engineering*



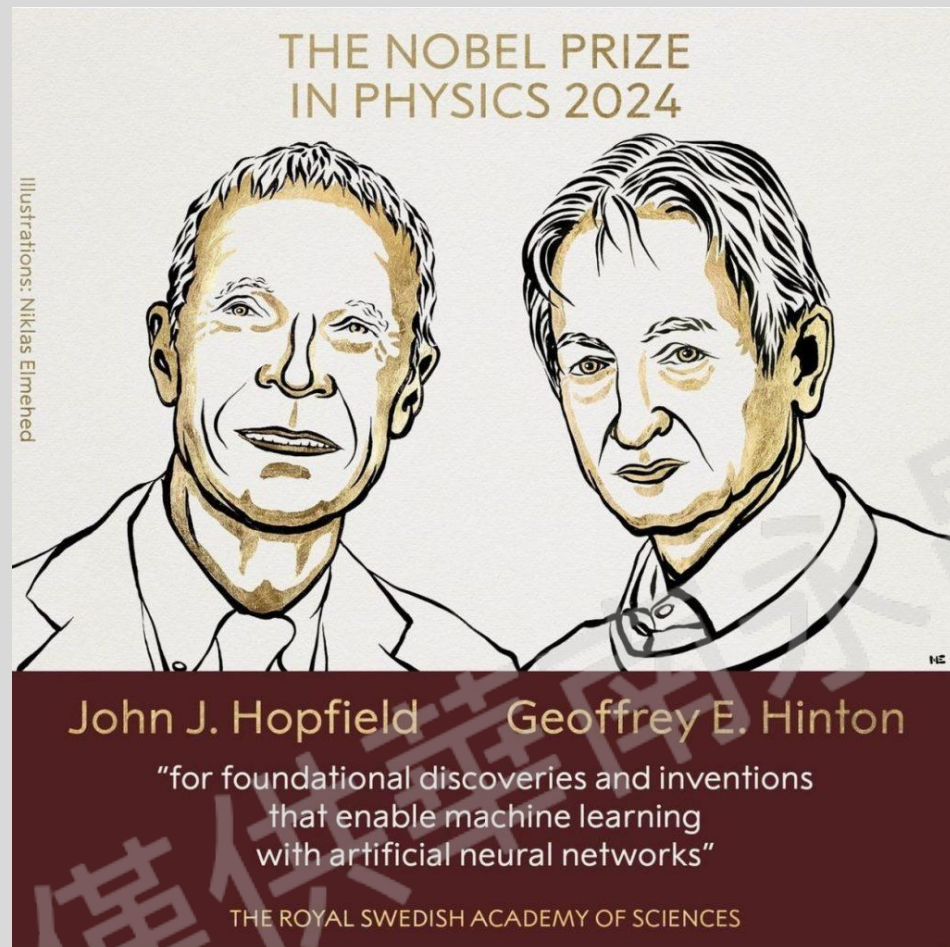
New Era of Nano-Photonics: *AI Journey through Quantum Implementation*

產業趨勢暨企業論壇/華南永昌證卷

May 22, 2026

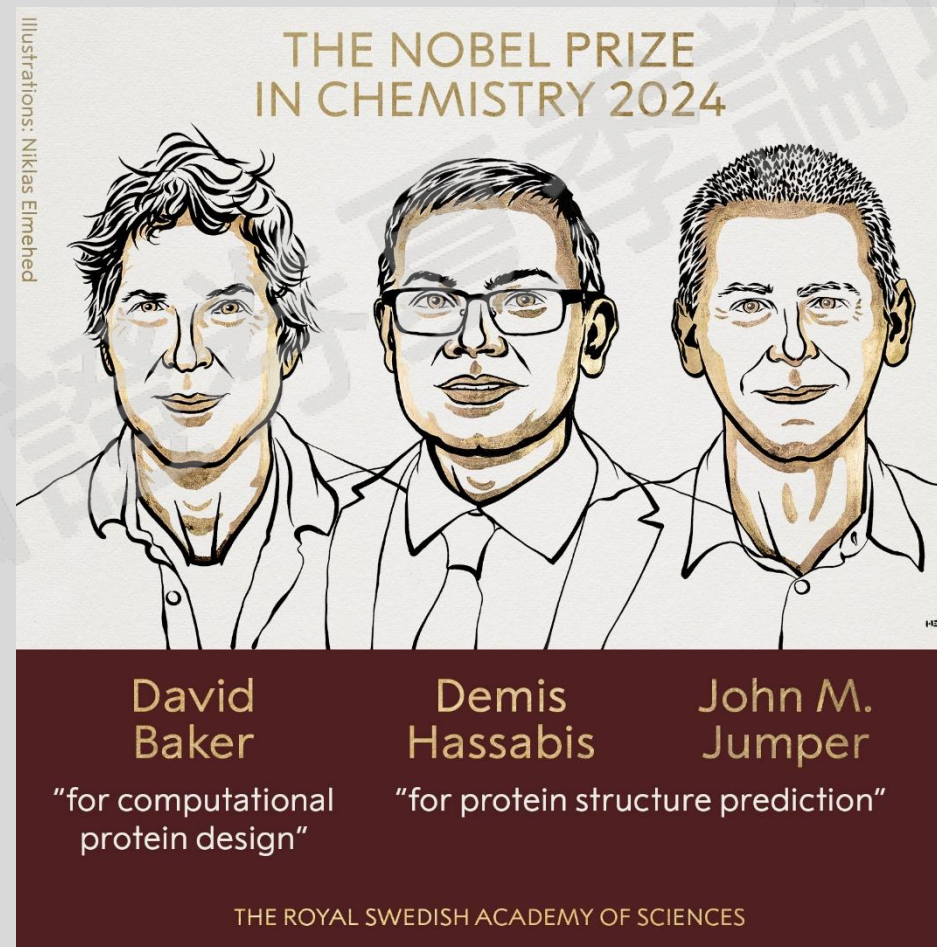
Redefining the Limits
Beyond

Nobel Prizes 2024



- Store and reconstruct the images.....
- Identify specific elements in pictures

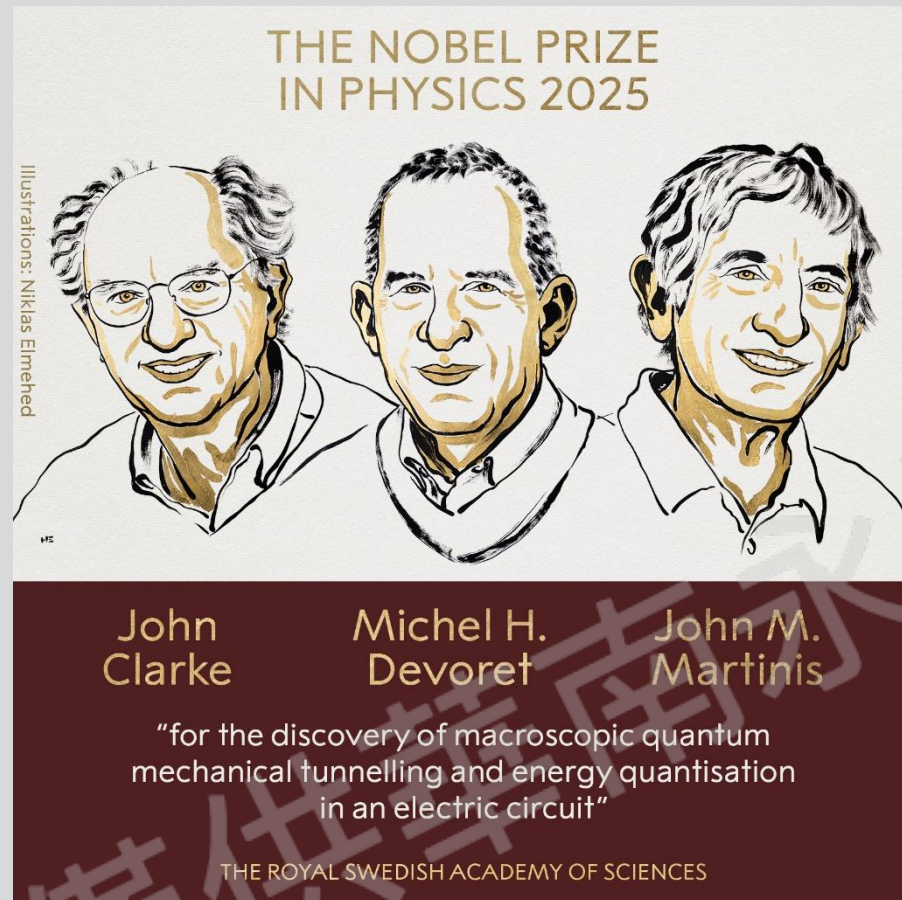
8 October 2024



- Entirely new kinds of proteins
- Predicting proteins' complex structures

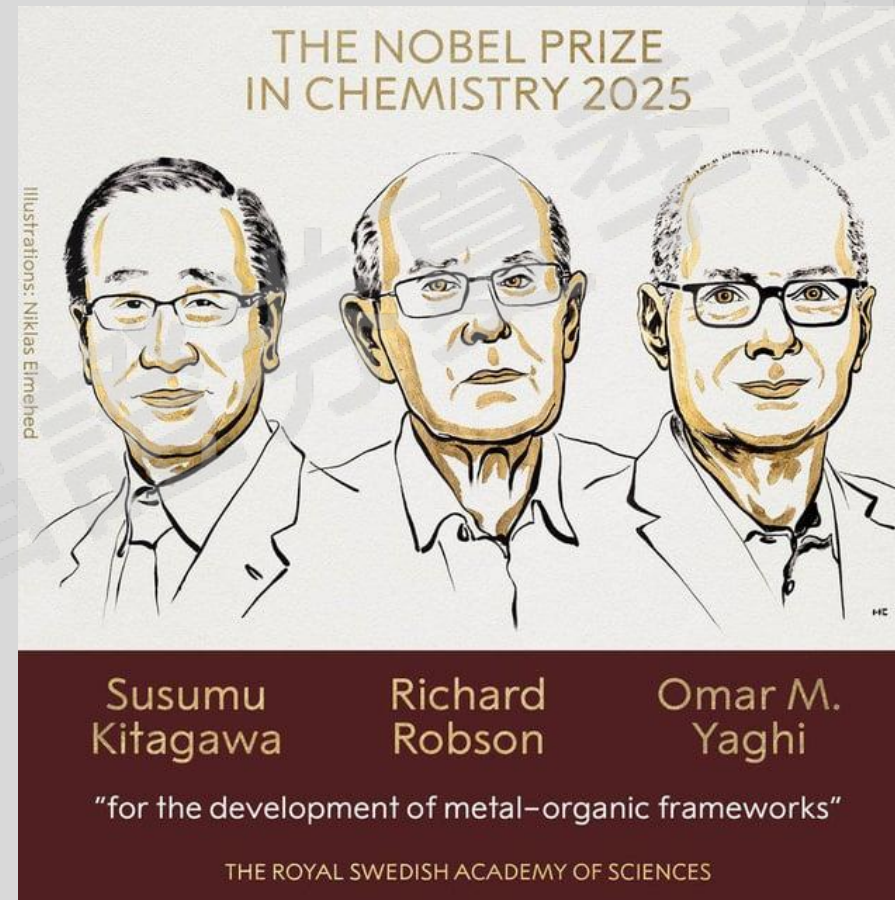
9 October 2024

Nobel Prizes 2025



7 October 2025

- Quantum mechanical properties on a macroscopic scale
- Basis for Quantum cryptography/computing/sensing



8 October 2025

- Harvest water, capture CO₂, store toxic gases
- Break down pharmaceuticals in the environment

“AI” Incubated in Data-Centers

cluster in Virginia, USA



AWS Data Center - 2026



Where power goes, data centers follow!



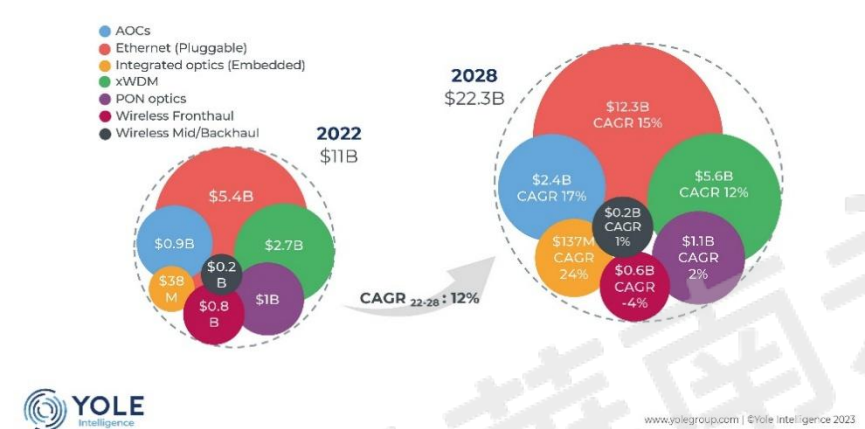
Small Modular Reactors (SMRs)

Market Capture

FC 30B by 2029

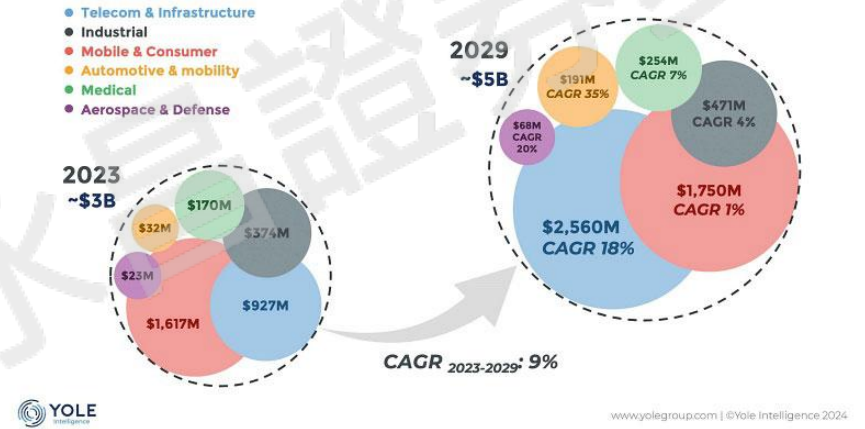
2022-2028 OPTICAL TRANSCEIVER REVENUE GROWTH FORECAST BY SEGMENT

Source: Optical Transceivers for Datacom and Telecom 2023 report, Yole Intelligence, 2023



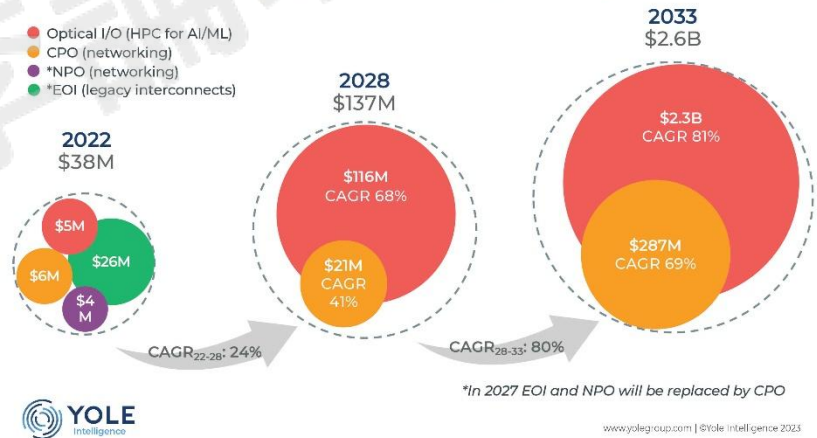
SEMICONDUCTOR LASER REVENUE FORECAST BY SEGMENT - 2023 VS. 2029

Source: Semiconductor Laser Industry report, Yole Intelligence, 2024



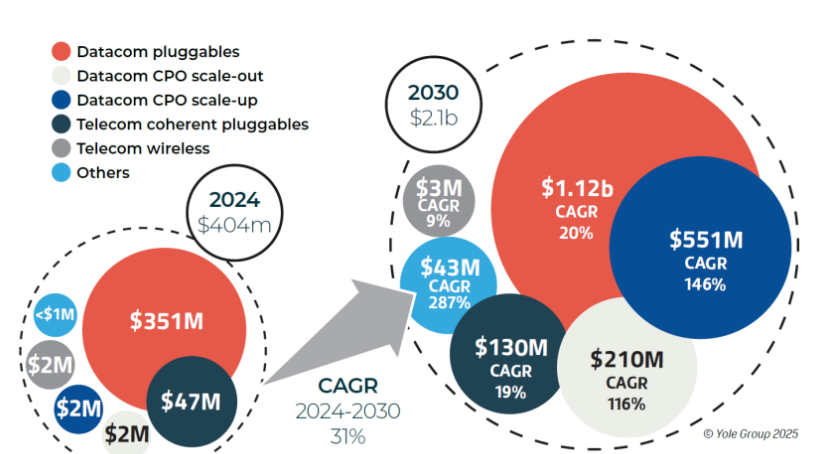
2022-2033 DATACOM OPTICS REVENUE FORECAST

Source: Co-Packaged Optics for Datacenter 2023 report, Yole Intelligence, 2023



Photonic integrated circuit market

Pics (dies) revenue growth forecast by technology platform (2024 vs. 2030)



high flying sectors

Datacom/Telecom

DFB, EML, CWL, Silicon Photonics

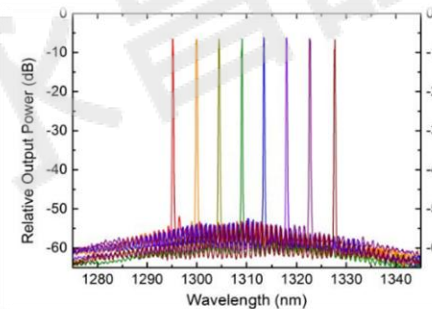
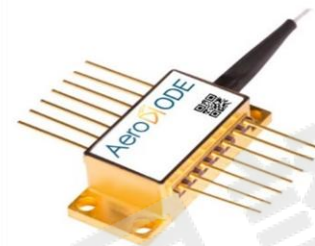
Datacentre for AI & ML

5G Connectivity Infrastructure



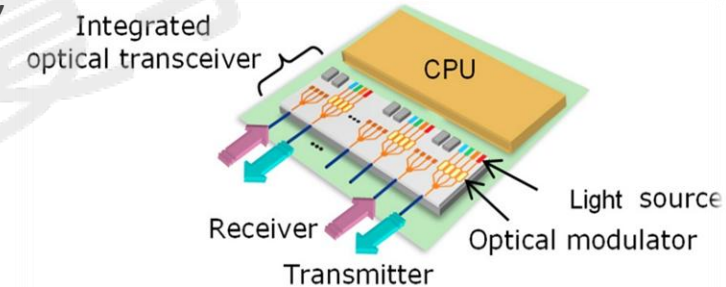
*High bandwidth
Multi-channels*

DFB/EML/CWL LD module



*High bandwidth
Power Efficiency
Small Footprint*

Optical transceiver architecture



*Silicon Photonics platform
Cost effective*

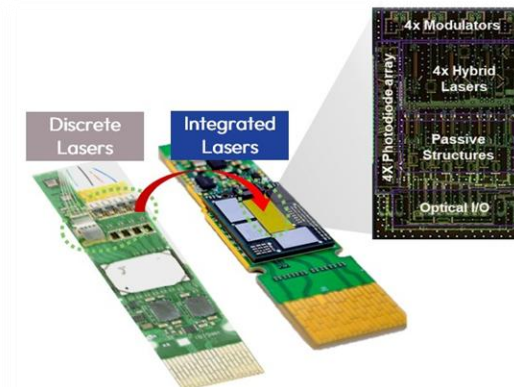
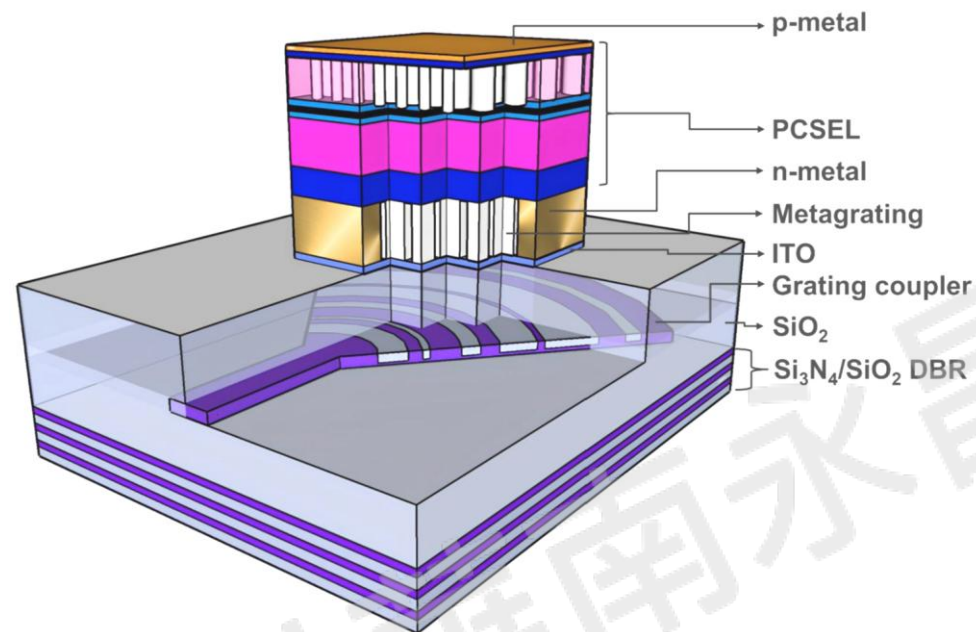


Image Credit: Getty Images

Co-Package Optics (CPO) Module

From silicon integrated photonics to quantum regime

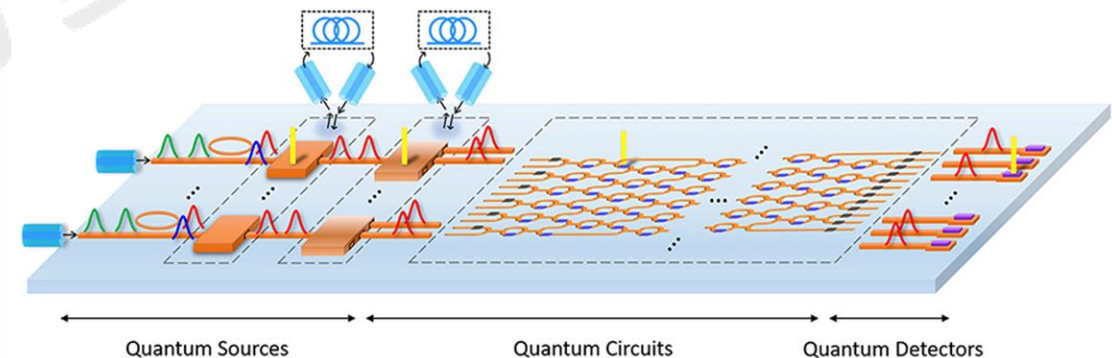


*Heterogenous integration
by single-photon sources*

- Efficient silicon photonics system by integrating PCSEL and metasurfaces

Journal of Lightwave Technology (2024)

- Room temperature *photonic quantum computing (PQC)* system



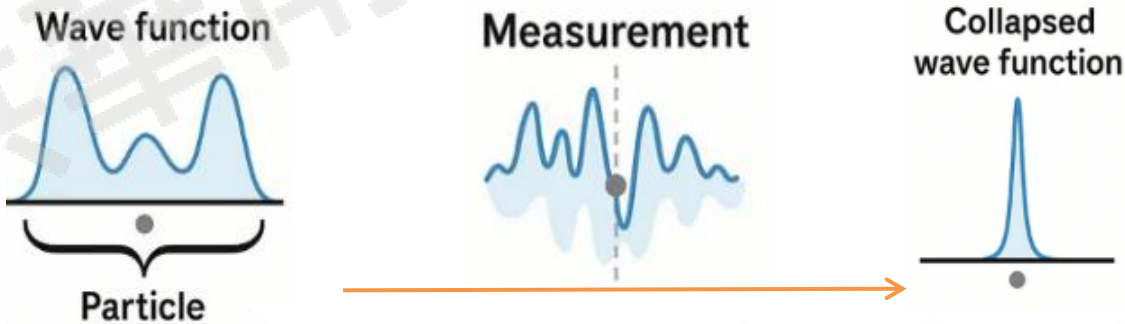
APL Quantum 1, 136123 (2024)

How the Quantum World Works?

Copenhagen Interpretation (哥本哈根詮釋)

概念	說明
❶ 波粒二象性 (Wave-Particle Duality)	微觀粒子既具有波動性也具有粒子性，哪一面被觀察到取決於「實驗方式」。
❷ 機率性 (Indeterminacy)	量子力學只能給出「發生機率」，而非確定結果。電子的位置與動量不能同時確定（海森堡不確定原理）。
❸ 波函數坍縮 (Wave Function Collapse)	系統在測量前由波函數描述（處於疊加狀態），但「一旦被測量」，波函數會立即坍縮到其中一個確定結果。
❹ 觀測者效應 (Observer Effect)	測量本身改變系統狀態。觀測行為不只是被動記錄，而是物理實在的一部分。
❺ 互補原理 (Complementarity)	波耳提出：波與粒、位置與動量等物理量是「互補」的觀點，不能同時被完整描述。只有實驗設定決定可觀察哪一面。

波函數坍縮





There was a time when the newspapers said that only twelve men understood the theory of relativity. I do not believe there ever was such a time... On the other hand, I think I can safely say that nobody understands quantum mechanics.

— Richard P. Feynman —

AZ QUOTES

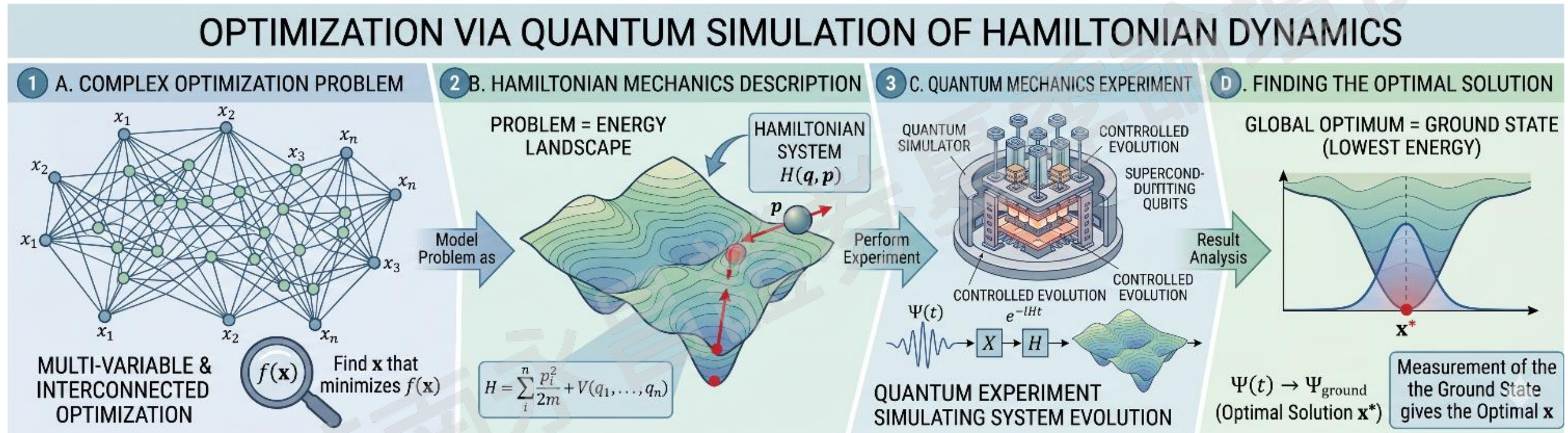


費因曼“詮釋”

Niels Bohr helped explain how atoms work and became one of the founders of quantum physics. Nobel Prize in 1922

Niels Bohr’s desk - 2023

What is Quantum Computing (simulation) capable of?



Step 1: Start with a hard optimization problem

You have many connected variables, and you want to find the best value of x that minimizes $f(x)$.

Step 2: Turn the problem into an energy landscape

The problem is modeled like a landscape with hills and valleys. Lower valleys represent better solutions.

Step 3: Describe it with a Hamiltonian

A Hamiltonian $H(q, p)$ represents the system's total energy.

Step 4: Simulate the system using quantum mechanics

A quantum computer or quantum simulator evolves the system over time.

Step 5: Find the lowest-energy state

The system moves toward the ground state, which is the lowest energy point.

Step 6: Read the optimal answer

The ground state gives the best solution, called x^* .

Different Quantum Hardware Paths



Main investment moves by NVIDIA:

Quantinuum — trapped-ion

NVIDIA's venture arm, NVentures, participated in Quantinuum's \$600 million funding round, valuing Quantinuum at about \$10 billion pre-money. Quantinuum is focused on trapped-ion hardware and **hybrid quantum-classical computing** with NVIDIA's CUDA-Q platform.

PsiQuantum — photons

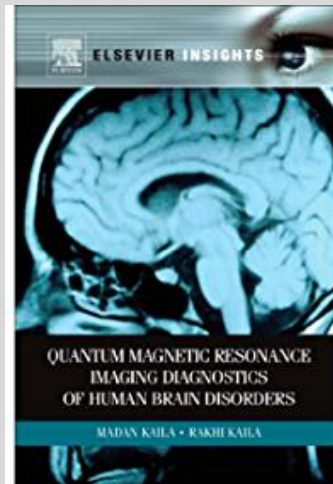
NVentures joined PsiQuantum's \$1 billion Series E round, which valued the company at about \$7 billion. PsiQuantum is building **photonic quantum computers** using light-based chips and is working with NVIDIA to integrate NVIDIA chips with its photonic systems.

QuEra — neutral-atom

NVIDIA also backed QuEra, a company using **neutral atoms** as qubits. This gives NVIDIA exposure to another major quantum hardware approach.

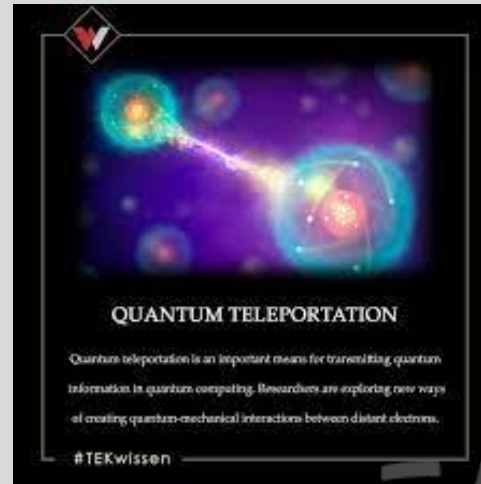
Potential Applications Implemented by Quantum Technologies

Quantum Imaging



Quantum **correlated** or **entangled photons** can extract image information from very weak, noisy, or indirect light signals.

Quantum Teleportation



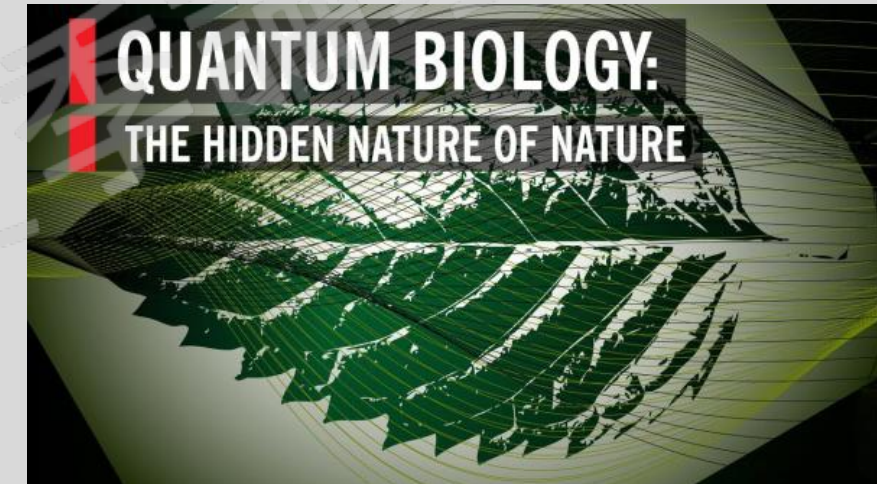
Entanglement plus classical communication transfers quantum information, not objects.

Quantum Navigation



Quantum **sensors at atomic energy levels** measure time, acceleration, rotation, gravity, or magnetic fields with extreme precision, allowing navigation even when satellite GPS is weak, jammed, or unavailable.

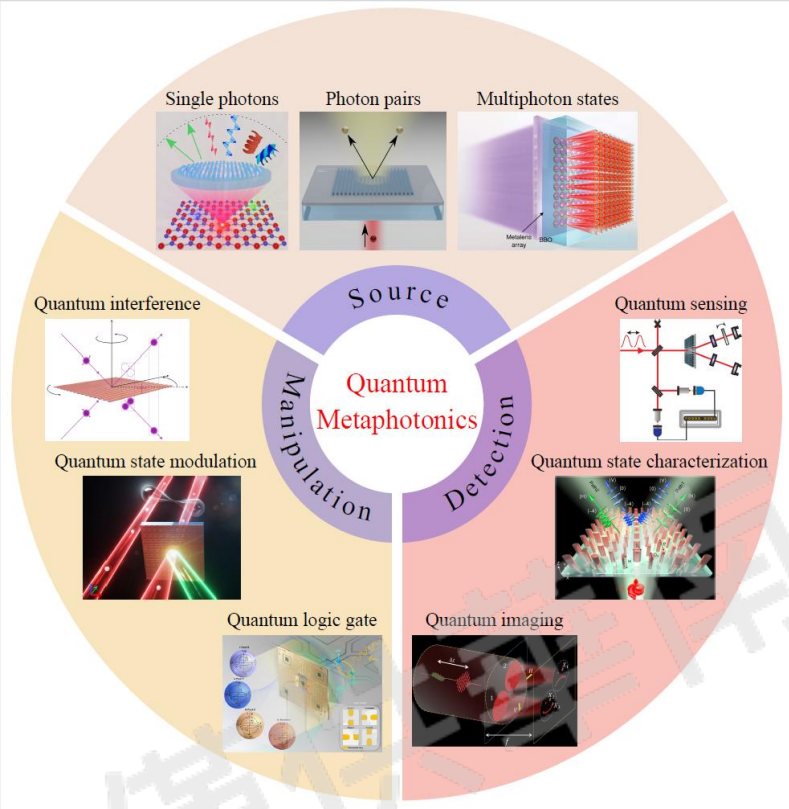
Quantum Biology



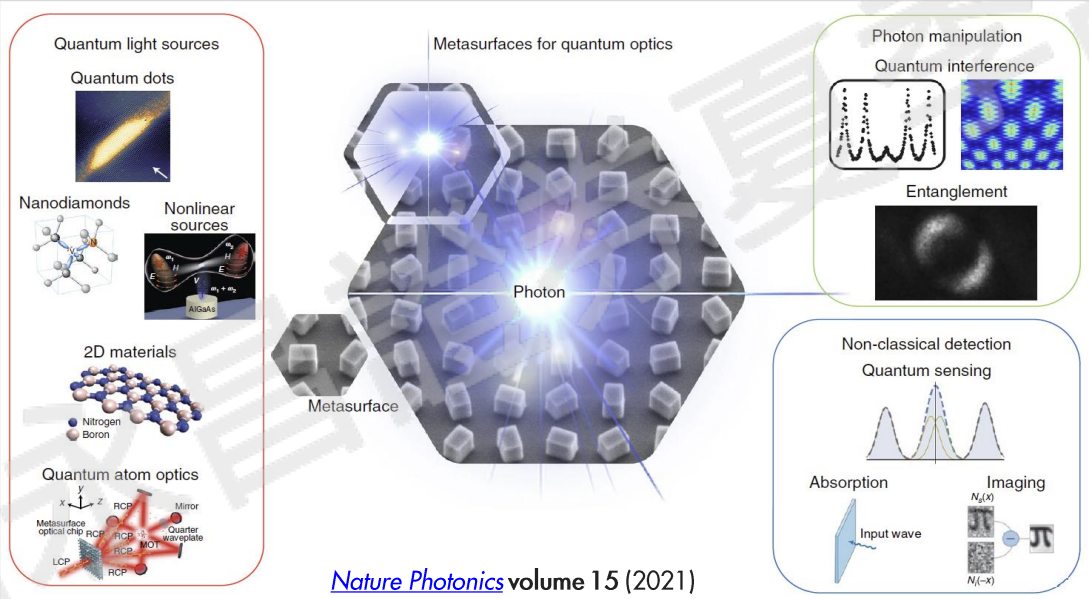
Enzyme reactions: particles such as protons or electrons may use **quantum tunneling** to cross energy barriers faster. **Photosynthesis:** energy may move through light-harvesting complexes using wave-like **quantum coherence**, though how biologically important this is remains debated. **Bird navigation:** some animals may sense Earth's magnetic field through **radical-pair spin** chemistry in proteins such as cryptochromes.

Metasurface

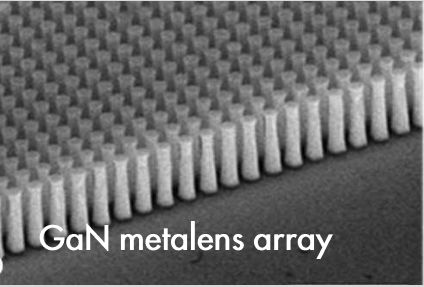
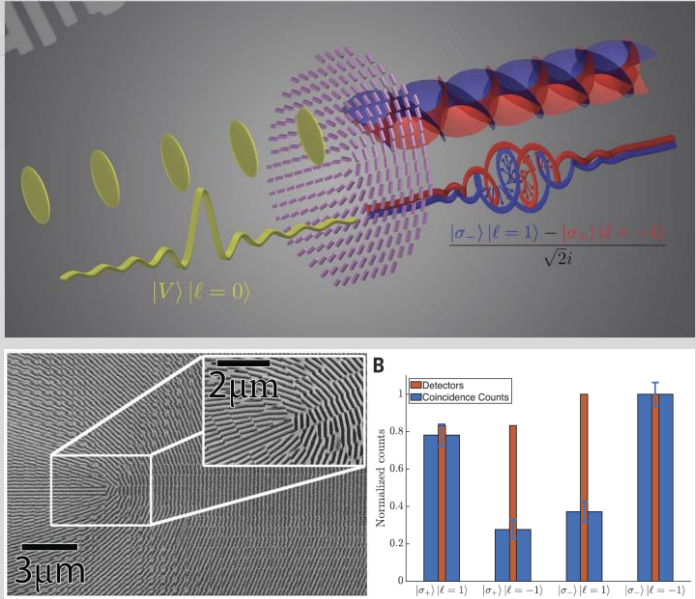
Quantum optics and photon manipulation



APL Quantum 1, 020902 (2024)

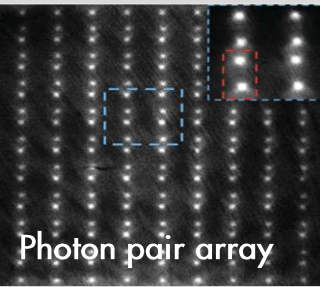


Nature Photonics volume 15 (2021)

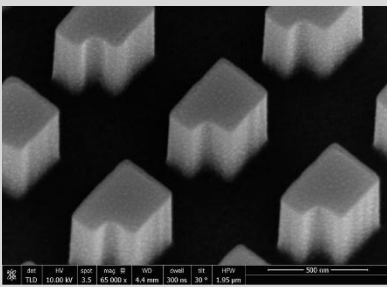


GaN metalens array

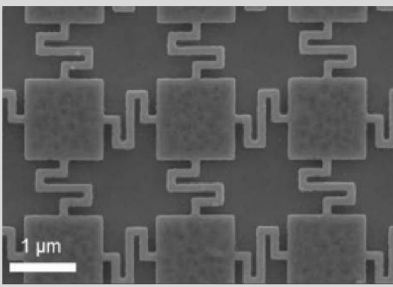
Science 368, 1487–1490 (2020)



Photon pair array



Science 377, 991–995 (2022)

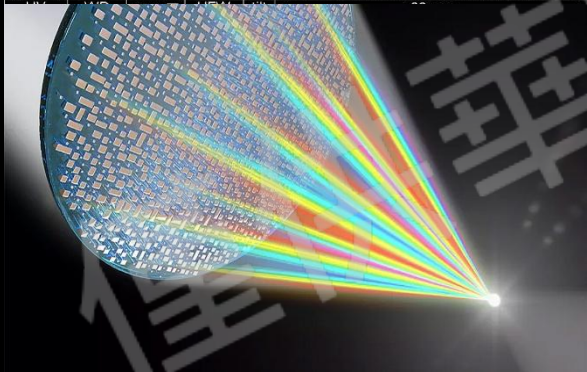
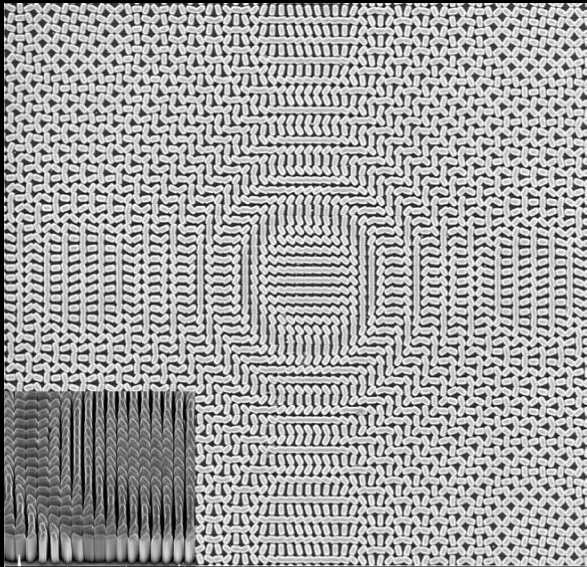


Optics Express Vol 26, 43598–43611 (2021)

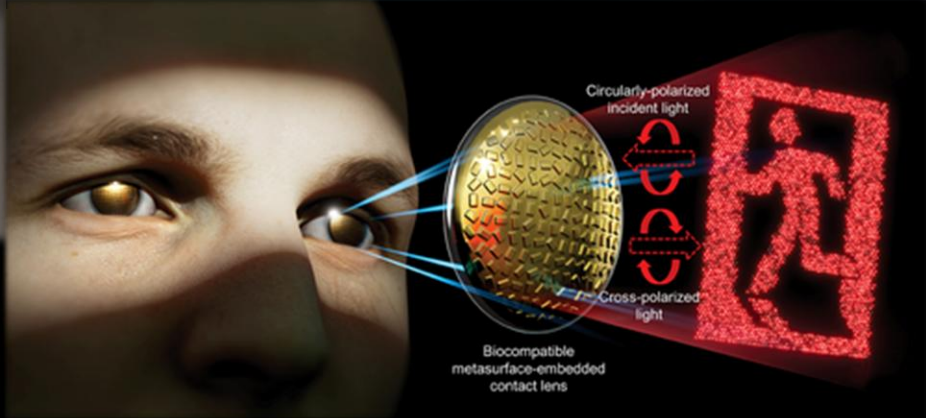
SCIENCE
Vol 361, Issue 6407
(2018)

Meta-Optics

Wavefront engineered optical plate



A metalens is restored to generate an output image that closely approximates the quality of the original "ground truth" image.

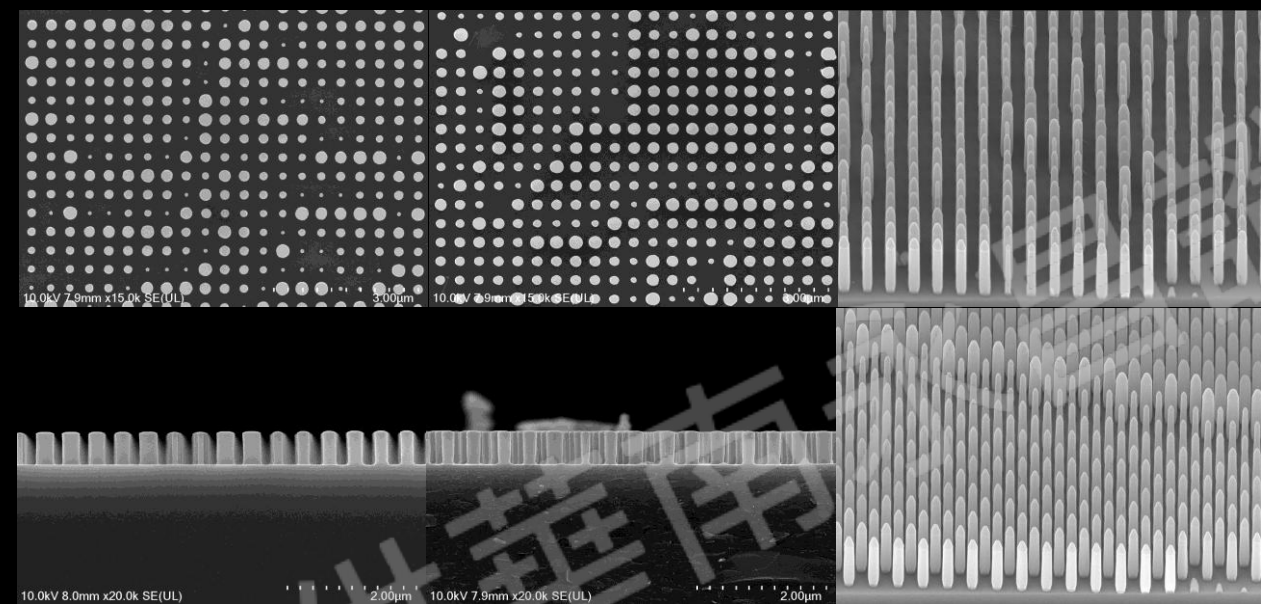


Adv. Sci. 2024, 11, 2407045

Spatial Computing by Photons

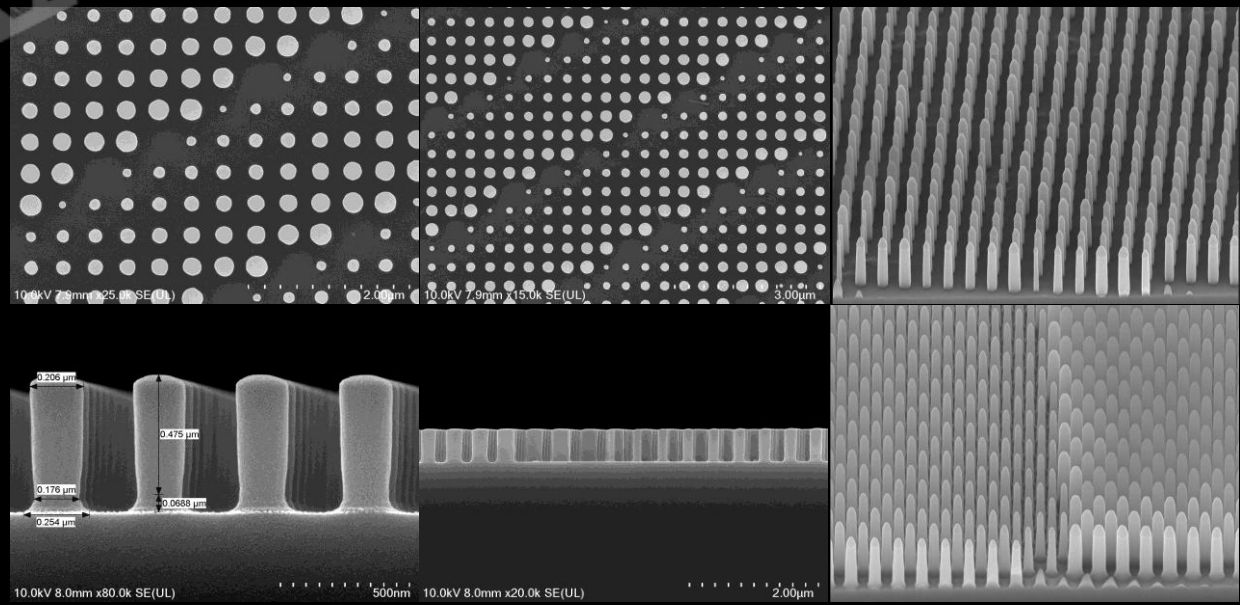
Metalens: Depth Time of Flight (IR dTOF) Sensor (a-Si, Glass)

TX Metalens



- Process Key Parameters
- Critical Dimensions: 60~320nm
 - Etch depth: 470~750 nm

RX Metalens

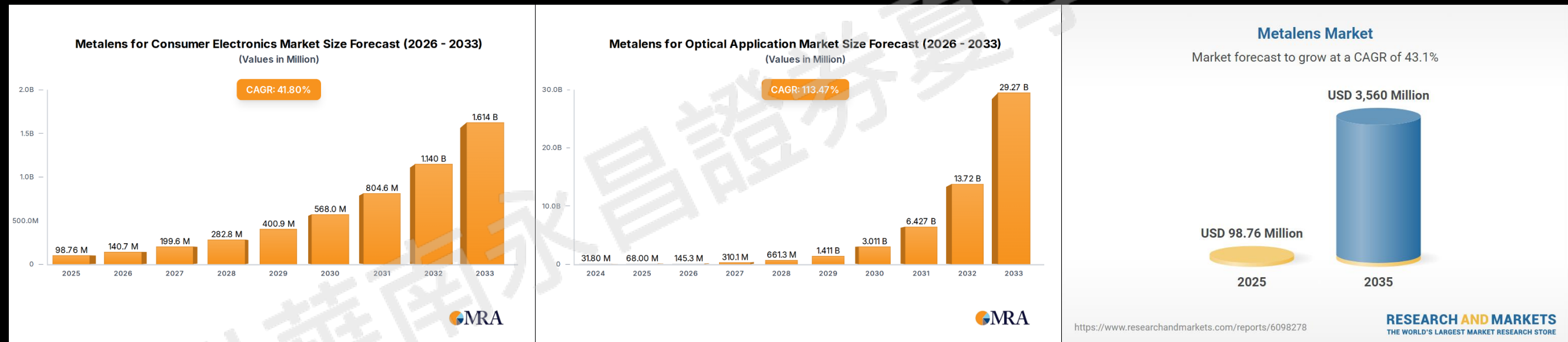


Structured Image



Market Capture

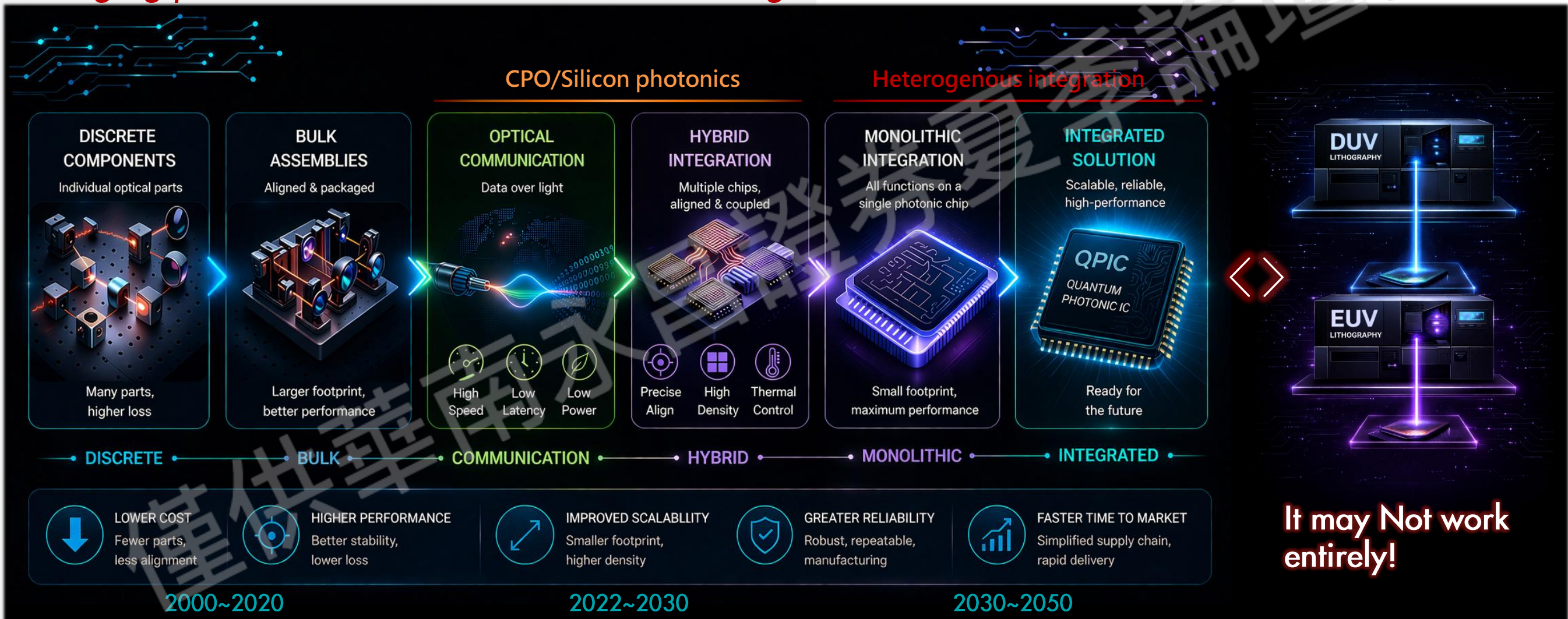
FC 30B by 2033



Ultra-high flying sector

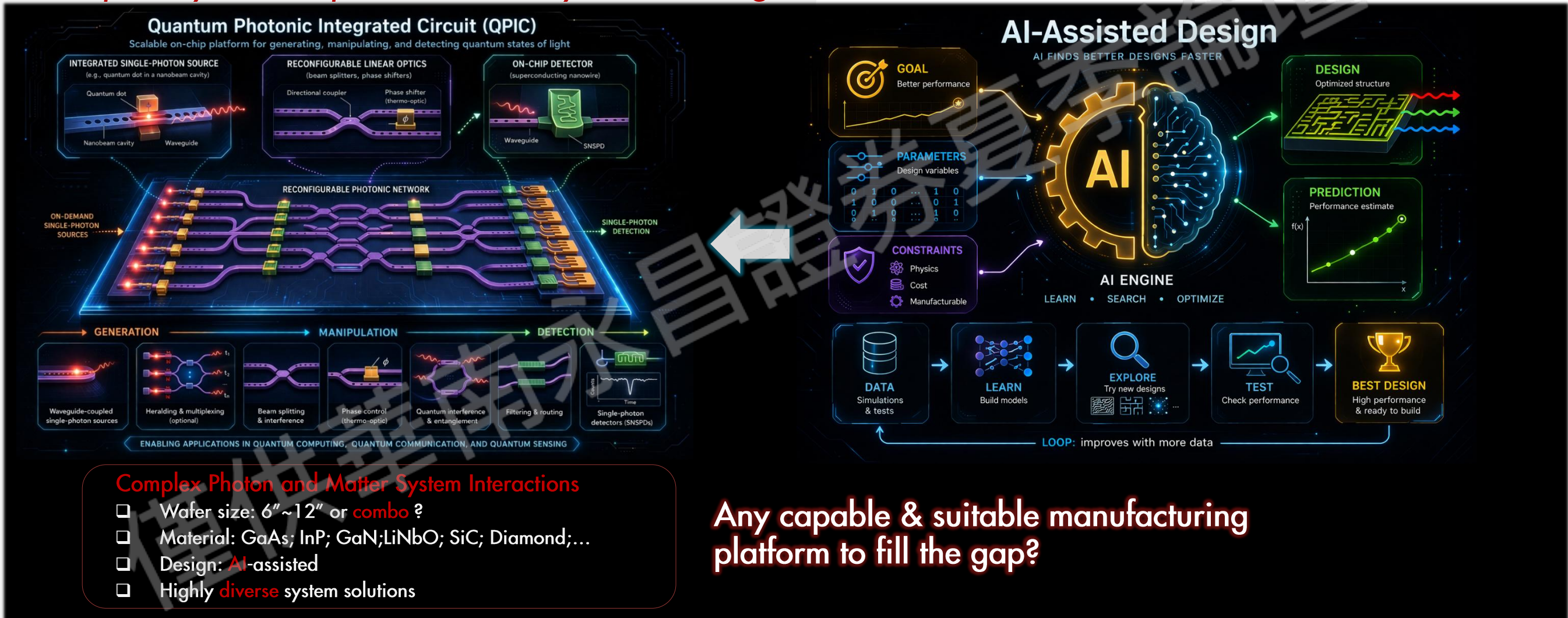
Manufacture Trend – from discrete to integrated solution

bringing photonics and data communication together



Existing Foundry Challenge – Heterogenous Integration

Complex system implementation by AI assisting



Emerging Processing Technology – Nanoimprint Lithography (NIL)

A low-carbon and cost-effective solution

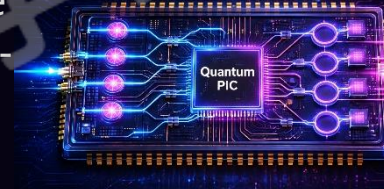
Traditional CMOS Scaling (Moore's Law)

Optimized exclusively for standard logic and memory on flat silicon.



More than IC Architectures

The massive convergence of AI-driven sensing, high-speed photonics, and quantum infrastructure.

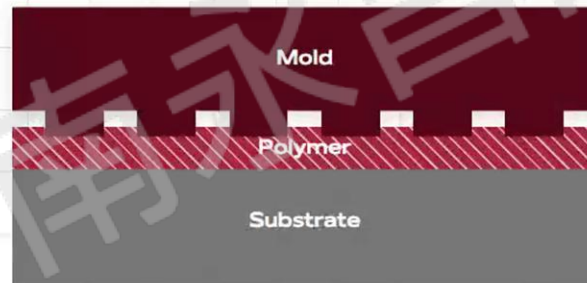


Press mold

UV/Heat curing

Release mold

Substrate etch



Nanoimprint lithography is a technique that enables the **replication of sub-micrometric fine features at wafer level** and can be applied onto any substrates. It is viewed as an alternative production solution compared with E-beam and EUV lithography.

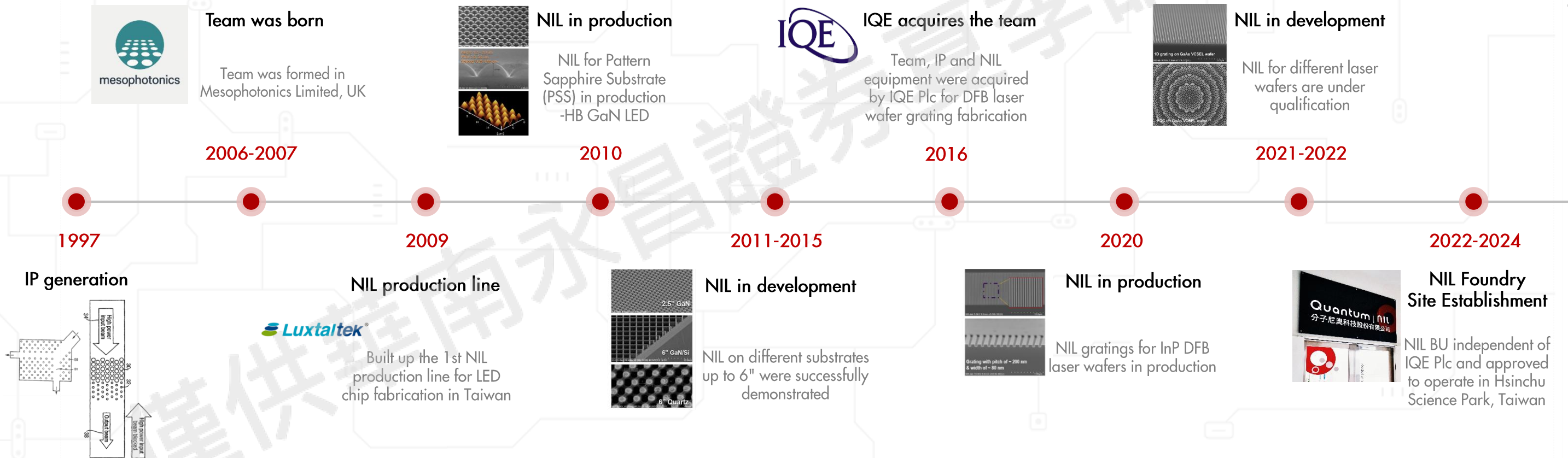
NIL is mainly used in a single stamp process for optical components. To widen NIL's applications, NIL shall be integrated with sophisticated processes including **multi-steps** with precision **alignments, material etch** and **photo-litho hybrid** steps.

IEEE Spectrum 2025:
Nanoimprint Lithography Aims to Take on EUV



Company Evolution

Pioneer in implementing NIL production



Flexible Hybrid Foundry Platform

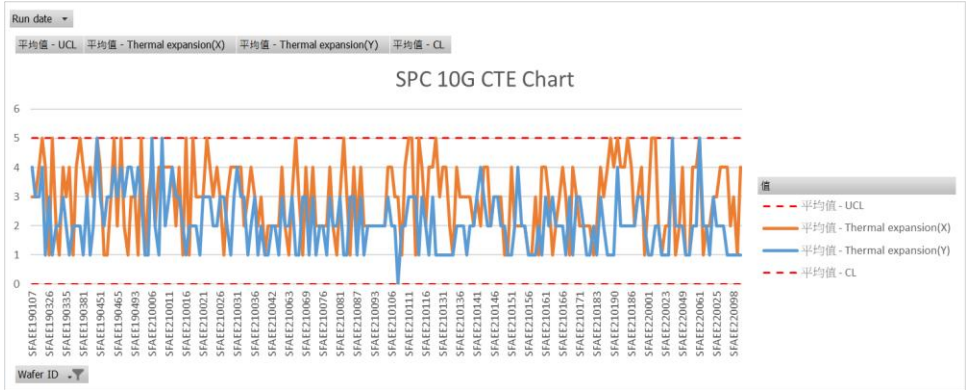
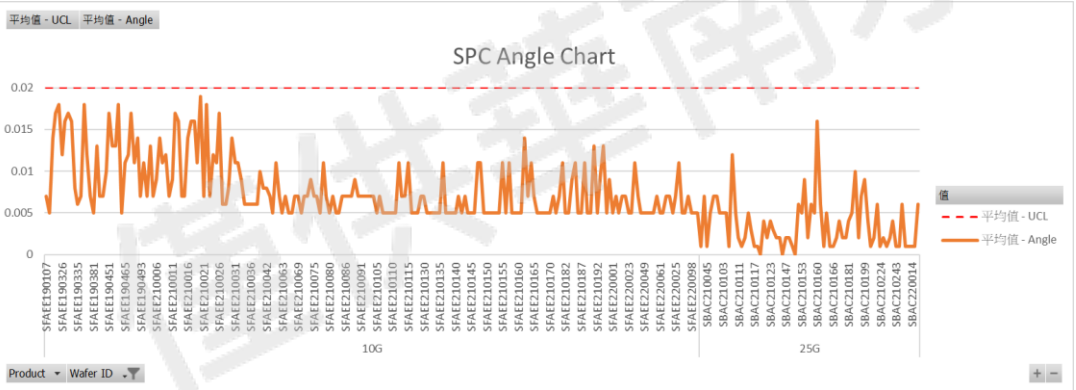
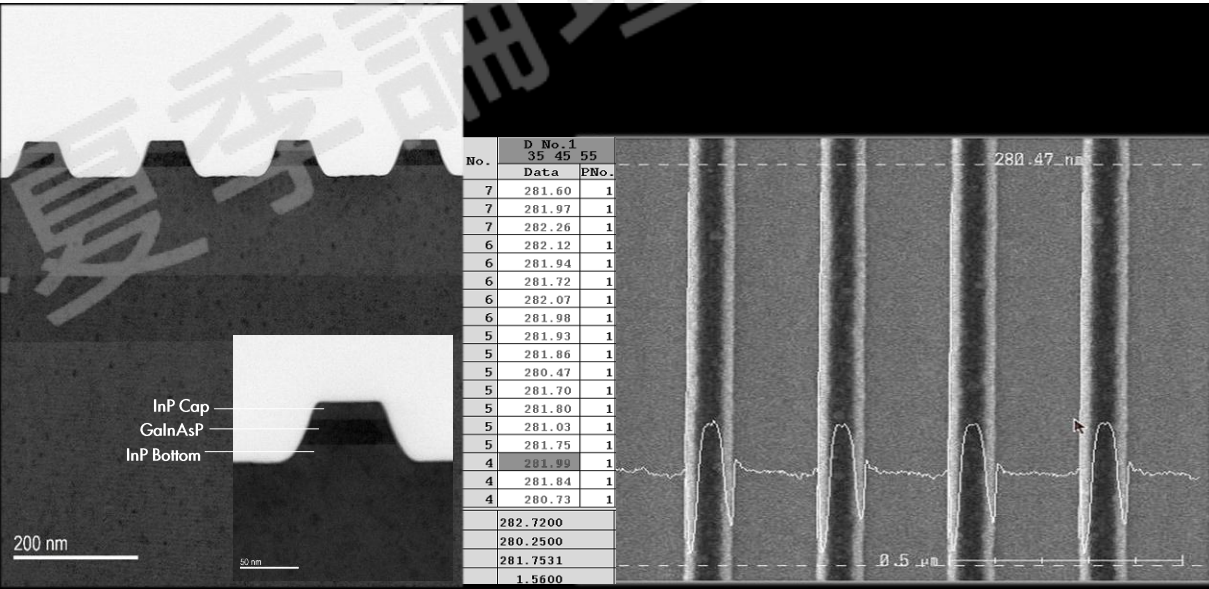
Hybrid NIL process builder

	Traditional DUV/EUV Platforms	Quantum NIL Flexible Hybrid Platform	Applied Market Characteristics • Structure resolution (CD) < 100 nm • Material function-oriented or non-Si substrate • Hybrid or complex structure – 2D/3D irregular distributed shape • Medium economic or customized production PM scale A complementary solution to the industry!
Primary Substrate	Standard Silicon (Si) Wafers	Non-Si, Fragile, and Bowed Substrates	
Minimum Feature Size	Optimized for CMOS Scaling	Down to 25 nm	
Material Versatility	Si mainly	High versatility (GaN, InP, Optical/Bio-materials)	
Process Economics	High CapEx resulting in Low Flexibility	Agile CapEx, From fast- prototyping to Scalable Production	
Primary Applications	Si-based Logic/Analog IC, Memory	Compound Semi, Metalenses, Quantum Optics, Biochips	

Laser Optics – a must for wavelength locking

Grating production for laser wafer (0.2nm accuracy)

- 10G/25G DFB, EML, CWL
 - > accumulate more than 1,500 wafers in production
 - > customer's device yield 85 ~ 90%
- Single NIL process to fabricate both of grating & process marks
 - > unique process flow to reduce the process steps for the customers
 - > etch depth varies from 10nm to 3 um
- Precise rotation angle control against the crystalline
 - > in-house customized NIL alignment tool < 0.02°
 - > offset control within wafer < +/-5 um

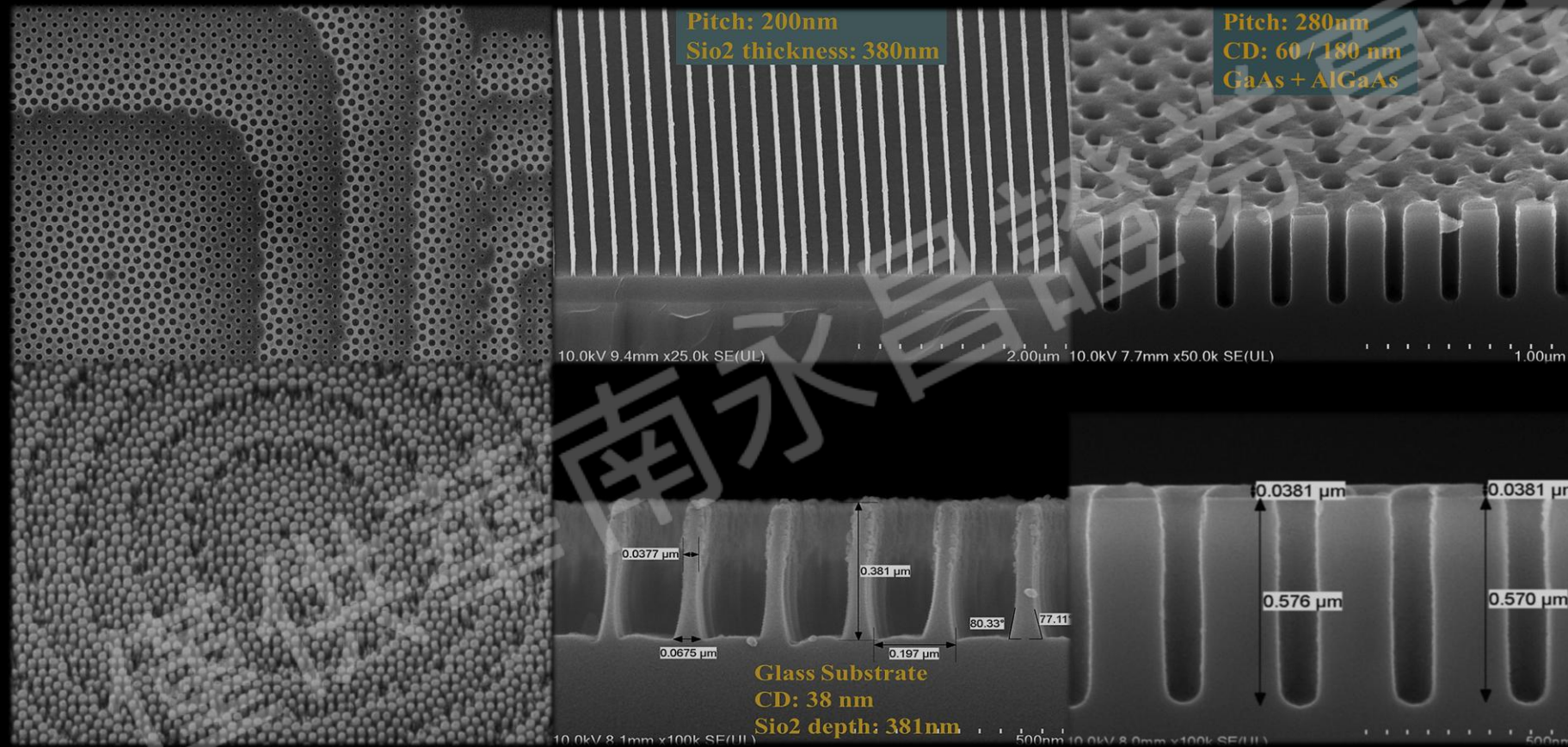


Process Key Parameters

- Critical Dimensions: 40~100nm
- Aspect Ratio: 0.2~1.5
- Materials: InP, GaAs, SiN
- Wafer Sizes: 2~6"

Meta-Optics

Metalens: Complex CD varying & high aspect ratio stacking



Process Key Parameters

- ❑ Critical Dimensions: 40~500nm
- ❑ Aspect Ratio: 5~10
- ❑ Materials: α-Si, III-V, TiO2, high-n Glass
- ❑ Wafer Sizes: 2~8"

Spatial Computing

Advanced Laser: PCSEL for extreme low/high power (GaAs, InP)

Progress in Photonic Crystal Lasers (PCSELs)

1999: Proposal and Demonstration of 2D Surface-Emitting Coherent Oscillation [5]

2005: Control of Polarization and Beam Shape [7,10]; Expansion of Wavelength (Blue violet) [12]

2010: 1D Beam Steering [13]

2015: Commercially Available (0.2W class) [25]; Watt-class Operation [14]; Single-lattice structure with right-angled triangular unit cell • Size: 200 μm x 200 μm

2020: Toward 10W-class Operation [16, this paper]

Process Key Parameters

- Critical Dimensions: 80/120nm
- Etch depth: 550nm

10.0kV 7.8mm x25.0k SE(UL) 2.00 μm

GaAs PCSEL

10.0kV 7.6mm x50.0k SE(UL) 1.00 μm

InP PCSEL

Various Applications of PCSELs for Society 5.0

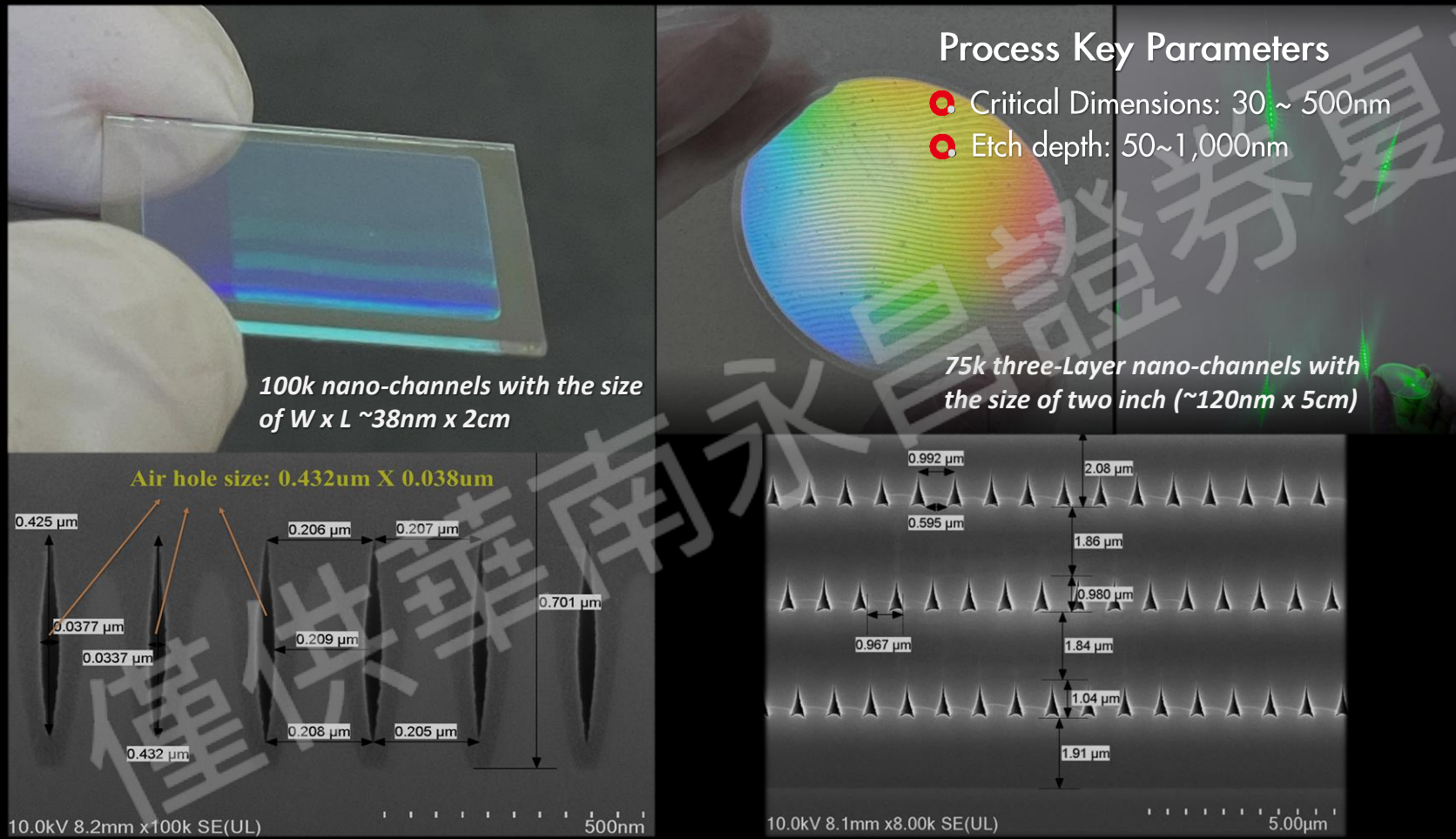
Smart Mobility (FA Robot), Smart Mobility (Farm Machines), Smart Mobility (Construction Machines), Smart Mobility (Automobile), LIDAR for Weather, Smart Processing (Electronics), Smart Processing (Automobile), Smart Processing (Solar Cell), Medical Application, Entertainment

Image

Object

Bio-sensing

Nano-channel Chiplets: Glass, SiN, SiO2

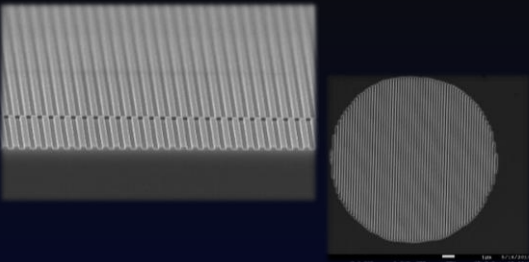


Market Opportunities

\$100B+ Deep-tech TAM

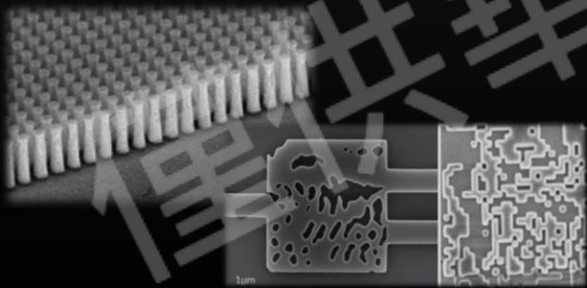
Data Center Optical Communications

\$20-30B transceiver market



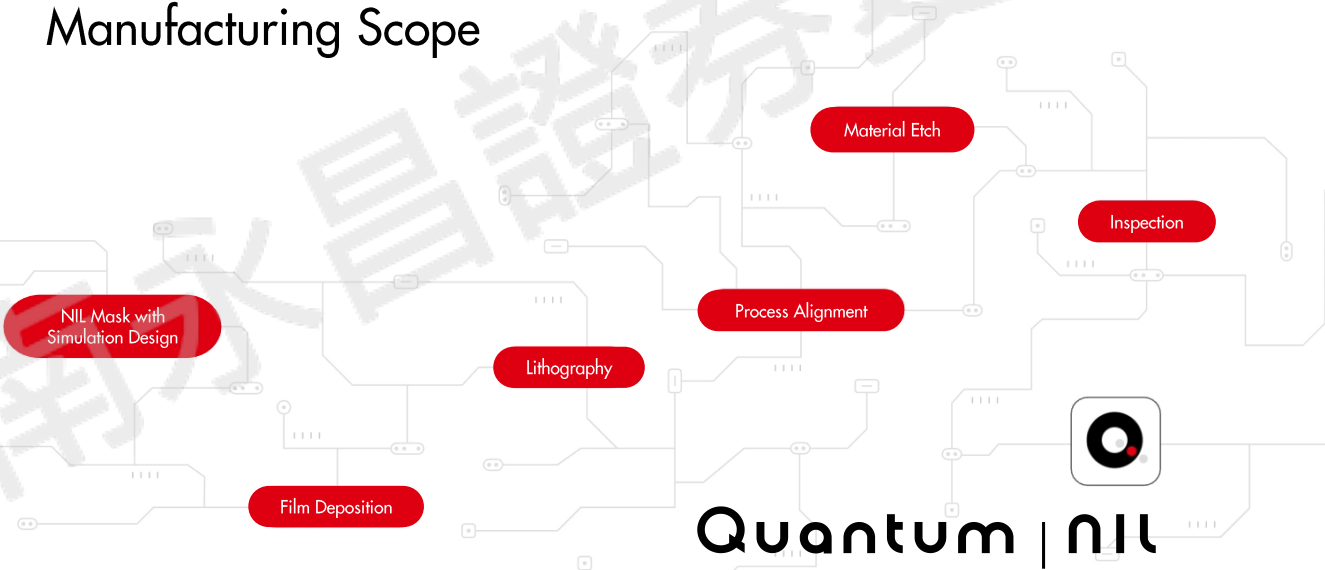
Quantum Technology

Projected to exceed \$100B globally



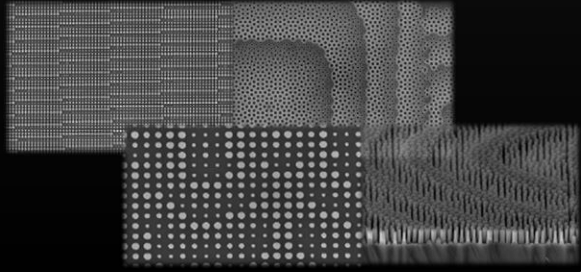
Foundry Business

Overview of Quantum NIL's Manufacturing Scope



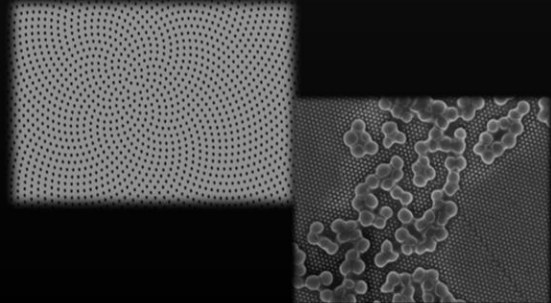
Meta-Optics for Spatial Computing & AR/VR

Multi-\$B emerging sector



Biotechnology & AI Sensing

Next-gen real-time diagnostics



Summary and Additional Highlights

- ❑ Nano-photonics plays a key role in next generation **AI data-center architectures**.
- ❑ **Hybrid Quantum computing** can accelerate AI training model generation while significantly **reducing the power** consumption.
- ❑ **Metasurfaces** integrated with hybrid materials can demonstrate multifunctional capabilities through **non-classical** optical characteristics.
- ❑ **Metalenses** offer a promising optical solution for **AR glasses** and serve as key components for **quantum optical systems**.
- ❑ Nanoimprint lithography (NIL) has evolved into a robust fine-structure fabrication technique compatible with standard semiconductor processes, enabling hybrid compound-semiconductor and optical-material integration for **quantum photonic integrated devices**.



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Miao-Li County 350401, Taiwan

TEL : 886-37-584-086 FAX : 886-37-584-225

Quantum | nll

分子 | 尼奧

Redefining the Limits
Beyond

ISO 9001:2015